

Prevalence of symptoms of depression in a nationwide sample of Korean adults

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Abstract

The prevalence and correlates of symptoms of depression in a nationwide sample of Korean adults, collected during the National Health and Health Behavior Examination Survey, were examined. A probability sample of 3,711 respondents (a response rate of 81.3%) completed the Center for Epidemiologic Studies Depression Scale (CES-D) and a variety of sociodemographic questions. In this sample 23.1% of males and 27.4% of females had scores above the cutoff point of 16 (probable depression) on the CES-D scale, and 6.8% of males and 10.4% of females were above the cutoff point of 25 (severe, definite depression). Apart from a few reports describing Afro-American and Puerto-Rican samples, these rates were somewhat higher than those found in the US and Western countries. In this report, female gender, fewer than 13 years of education, and disrupted marriage (widowed/divorced/separated) proved to be statistically significant predictors of severe, definite symptoms of depression. © 1998 Elsevier Science Ireland Ltd. All rights reserved.

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1. Introduction

Depressive disorder is one of the most common mental disorders and is considered a major public health problem in the United States and some

European countries (Weissman et al., 1981; Regier et al., 1988). Surveys of the prevalence of depression and depressive symptoms among people residing in the community have yielded widely differing estimates (Stallones et al., 1990). Case ascertainment in psychiatric research has been accomplished using one of two approaches: diagnostic or non-diagnostic (Roberts et al., 1990). Non-diagnostic methods have traditionally used

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brief screening instruments to determine the prevalence and severity of depressive symptoms in the community (Blumenthal, 1975; Comstock and Helsing, 1976; Weissman et al., 1977; Radloff and Locke, 1986). Reported prevalence of significant depressive symptomatology has ranged from 15 to 44%, and estimates of the prevalence of depressive symptoms have been somewhat higher than those of clinical depression by the diagnostic method (Barnes et al., 1988; Potter et al., 1995).

The Center for Epidemiologic Studies Depression Scale (CES-D), a self-reporting depression scale, has been widely used in population surveys in the US and over the world (Radloff, 1977), and satisfactory levels of its reliability and validity have been reported (Radloff, 1977; Myers and Weissman, 1980; Eaton and Kessler, 1981; Roberts and Vernon, 1983; Weissman et al., 1986). In studies relying on the CES-D, estimates of the prevalence of depression have been somewhat higher than those based on diagnostic measures, such as Diagnostic Interview Schedule (DIS) and Present State Examination, but have still been fairly consistent. Most studies have reported a prevalence of between 10 and 28%. The CES-D has been translated into Korean, and its reliability and validity have been confirmed (Cho and Kim, 1993). In this report, the authors have suggested two optimal cutoff points: one is the universal cutoff point of 16, which most effectively detects and covers 'probable' depressive symptoms; the other is 25, which best corresponds to DSM-III-R major depression and is regarded as the cutoff point indicating severe, 'definite' depression symptoms. The CES-D has also been applied in several surveys to other ethnic groups (Vernon and Roberts, 1982; Vega et al., 1984; Gary, 1985; Shima et al., 1985; Burnam et al., 1987; Rankin and Galbraith, 1993; Potter et al., 1995). In some such groups, however, other cutoff points have been suggested (Husaini et al., 1980; Weissman et al., 1986; Parikh et al., 1988), and particular emphasis has been placed on the need for a higher cutoff point in Asian countries and in some Hispanic ethnic groups (Cho et al., 1993; Iwata et al., 1994; Cho and Kim, 1998); the need

for a higher cutoff point is due to the different response styles to positive affect items.

To date, only one nationwide community survey has reported the prevalence of depression in Korea (Lee et al., 1990a,b); the DIS was used to measure DSM-III major depression and dysthymia. No nationwide community-based study employing a depression scale has, however, been carried out to determine the prevalence of symptoms of depression, and no analysis of depressive symptoms has included the independent contributions of personal attributes. The present study analyzes symptoms of depression, as measured by the CES-D, among a nationwide probability sample of adults aged between 20 and 59; data were collected between June and August 1995, through the National Health and Health Behavior Examination Survey (NHHBES) by the Korean Institute for Health and Social Affairs. In addition to estimating the prevalence of depressive symptoms in Korea and comparing the rate with those in other countries, we analyzed predictors of these symptoms.

2. Methods

2.1. Sample and sampling procedures

The sampling frame was based on the 1994 National Census Registry and was considered representative of the adult population aged between 15 and 69 years, a probability sample of the non-institutionalized population of Korea. All 12 457 570 households in Korea were divided into 185 149 primary sampling units (PSUs), each of which contained 40–90 (mean 67.2) households. Institutions and small islands, the population of which was less than a half of 1% of the total national population, were not included. Stratification was made on urban and rural areas. Because the ratio of the number of households between urban and rural areas was 77:23, PSUs in rural areas were oversampled by a factor of two. From the total number of primary sampling units, 110 were randomly selected for this survey; 69 were from urban and 41 from rural areas. Each PSU comprised, on average, approximately 70

households. All 15–69-year-old members of each selected household were first enumerated and listed according to sex and age. Within each household, one eligible person was randomly selected as the respondent in an interview lasting one hour; the probability of being chosen was therefore inversely proportional to the size of the household. The NHHBES was finally based on a sample consisting of 6,480 persons aged 15–69. The CES-D was administered to adults aged 20 and over, but because the response rate of those aged 60 or over was less than 50%, our analysis focused on subjects aged under 60 ($N = 4,563$).

2.2. Interview and data collection

A total of 93 interviewers, most of whom had worked for many years on national surveys of this kind, were hired and trained to conduct the interview. Data were gathered between June and August, 1995. Of the 4,563 adults aged 20–59 who were sampled, 3,711 respondents completed the household interview, a response rate of 81.3%; those who were unable to respond to five or more items on the CES-D were not included. On the basis of information included in the 1994 census, respondents were compared with the whole national population; regard to a number of important sociodemographic characteristics, the respondents were reasonably representative of Korea.

Significant depressive symptomatology was measured by the CES-D, which was used after demographic questions and prior to other medical questions. This is a 20-item scale originally developed to assess current depressive symptomatology in community settings (Radloff, 1977). The Korean language version of the CES-D has been used with acceptable results (Cho and Kim, 1993, 1998). It can be self-administered, and responses to the items are made on a four-point scale, from zero (rarely or none of the time) to three (most or all of the time), and refer to how often the respondent has felt the item during the previous week. A cutoff score of 16 or above is conventionally used to indicate significant current depressive symptomatology, or caseness (Radloff and Locke, 1986). Cho and Kim (1993) suggested that a cutoff point of 25, as determined by the Korean

version of the CES-D, best predicted severe symptoms of depression (definite caseness) among Koreans.

2.3. Statistics

All adults aged between 20 and 59 years were included in the analyses. A response was defined as valid if 16 or more of the 20 items of the CES-D were completed, and unweighted and weighted distributions of the sociodemographic characteristics of respondents were obtained (Table 1).

Weighted prevalences of CES-D caseness ('probable' caseness, CES-D score of 16 or more; 'definite' caseness, CES-D score of 25 or more) were calculated, and for these, 99% confidence intervals were computed (Table 2). Stratification sometimes led to a small number of cells, and so to enhance the stability of the estimates, some variables were collapsed. To assess their independent contributions to the presence or absence of definite CES-D caseness (score of 25), sociodemographic characteristics were entered into logistic regression models (Table 3). Regression was calculated using the SAS LOGIST procedure with unweighted data. Variables entered into the model were such that they were comparable with domains examined in previous epidemiological studies of depression (Narrow et al., 1990). Because of the large number of coefficients estimated, Bonferroni's procedure was used to compute P -values (Woolson, 1987). In the logistic regression equation for CES-D caseness, 11 coefficients were estimated. To achieve 95% confidence that a coefficient did not reach significance by chance alone, a P -value of less than 0.0045 (0.05/11) was employed in significance testing.

3. Results

Of the 4,563 adults aged 20–59 years who were sampled, 3,857 completed the household interview; of these, 3,711 persons (81.3% of the initial sample) completed the CES-D, and their demographic characteristics are shown in Table 1. It can be seen that respondents were mostly married, with a spouse living in the household; over

Table 1

Sociodemographic characteristics (% distribution) of Korean adults 20–59 years who responded to CES-D, Korean NHHBES 1995

Characteristic	CES-D completed		
	<i>n</i>	Unweighted %	Weighted %
Area	3711		
Urban	2641	71.2	84.4
Rural	1070	28.8	15.6
Sex	3711		
Male	1696	45.7	48.5
Female	2015	54.3	51.5
Age	3711		
20–29	1001	27.0	30.0
30–39	1349	36.4	31.5
40–49	817	22.0	23.4
50–59	544	14.7	15.1
Education	3709		
0	123	3.3	2.5
1–6	522	14.1	12.7
7–9	612	16.5	15.7
10–12	1565	42.2	42.4
13–16	826	22.3	24.8
17 +	61	1.6	1.8
Marital status	3711		
Never married	720	19.4	22.4
Married	2751	74.17	73.4
Widowed	137	3.7	2.7
Divorced	64	1.7	1.0
Separated	39	1.1	0.5
Occupation	3711		
High officer	20	0.5	0.7
Professional	149	4.0	4.0
Semi-professional	123	3.3	3.8
Office worker	380	10.2	10.9
Service worker	583	15.7	15.5
Farmer or fisherman	408	11.0	6.6
Licensed worker	373	10.1	11.0
Machine operator	166	4.5	4.9
Laborer	265	7.1	7.2
Soldier	17	0.5	0.5
Student	136	3.7	5.3
Housewife	902	24.3	24.4
No occupation	189	5.1	5.3
Self-assessment of his/ her living standard	3710		
Very low	142	3.8	3.7
Low	900	24.3	24.5
Average	2403	64.8	64.4
High	251	6.8	6.9
Very high	14	0.4	0.5

Table 1 (continued)

Characteristic	CES-D completed		
	<i>n</i>	Unweighted %	Weighted %
Monthly income (unit: million won) ^a	3681		
0–0.5	387	10.5	7.0
0.51–1.0	1069	29.0	25.6
1.01–1.5	1021	27.7	27.8
1.51–2.0	650	17.7	19.7
2.01–2.5	255	6.9	8.4
2.51 +	299	8.1	11.4
Social class	3633		
High	462	12.7	14.8
Upper middle	1047	28.83	30.0
Lower middle	1388	38.2	36.8
Low	736	20.3	18.3

^a One million won was equivalent to US\$ 1,218 at the time of this survey.

one-quarter had at least some post-high school education. Occupations were listed according to the standard classification of the Korean National Census Registry, and taking five factors (educational level, monthly income, occupation, job status, and subjective assessment of his/her living standards) into account, four social classes (high, upper-middle, lower-middle, and low) were defined. Table 2 shows mean CES-D scores, standard errors, and caseness rates by two cutoff points (16 and over, probable caseness; 25 and over, definite caseness), with 99% confidence intervals. The mean CES-D score was 10.57 (male, 10.01; female, 11.10), and the probable caseness rate for the entire sample was 25.31%; that of definite severe depression, as defined by a cutoff score 25 and over, was 8.68%. Women had significantly higher CES-D caseness rates than men. Respondents with a lower educational level also had significantly higher rates than those whose level was higher. Respondents whose marriages had been disrupted also had significantly higher rates than respondents in intact marriages and those who had never been married. Currently unemployed respondents had significantly higher rates than those who were employed. Persons earning less than half a million won (equivalent

of US\$ 609 at the time of this survey) a month had rates more than twice as high as any of the other income categories, although this difference did not reach statistical significance. Respondents whose social class was low also had significantly higher rates than those at higher levels. As for age, those in their twenties showed the highest rate of depressive symptoms.

Table 3 shows the results of the logistic regression model in which the presence or absence of CES-D caseness of definite depression (≥ 25) was regressed on sociodemographic variables, which were collapsed in order to enhance the stability of the estimate. Women were significantly more likely to have high levels of depressive symptoms than men ($P = 0.0023$). Groups with below 6 ($P = 0.0022$) and below 12 years of education ($P = 0.002$) had high levels of depressive symptoms when compared to respondents educated for 13 years or over. Those with disrupted marital status were significantly more likely to have high levels of depressive symptoms than those in an intact marriage ($P = 0.0001$). Unemployment and low income (less than 0.5 million won a month) showed higher odds ratios, but in this model did not reach statistical significance.

Table 2

CES-D scores and probable and definite caseness rates (weighted) per 100 by sociodemographic characteristics, Korean NHHBES. (N = 3711)

Characteristic	CES-D score		Caseness rate (probable) ^a		Caseness rate (definite) ^b	
	Mean	S.E.	Rate	99% confidence limits	Rate	99% confidence limits
Total	10.57	0.14	25.31	(23.65–27.04)	8.68	(7.64–9.85)
Area						
Urban	10.67	0.15	25.81	(24.00–27.72)	8.69	(7.56–9.97)
Rural	10.05	0.35	22.56	(18.66–27.00)	8.59	(6.15–11.84)
Sex						
Male	10.01	0.19	23.07	(20.79–25.52)	6.83	(5.53–8.40)
Female	11.10	0.21	27.42	(25.05–29.92)	10.42	(8.86–12.21)
Age						
20–29	11.73	0.26	30.96	(27.75–34.37)	10.07	(8.10–12.45)
30–39	10.01	0.24	22.29	(19.52–25.34)	8.08	(6.36–10.21)
40–49	9.64	0.27	20.87	(17.76–24.37)	7.57	(5.67–10.03)
50–59	10.89	0.38	27.26	(22.98–32.00)	8.89	(6.37–12.24)
Education						
0	13.92	1.22	35.71	(24.76–48.32)	15.18	(8.09–26.35)
1–6	11.36	0.43	26.93	(22.30–32.12)	12.75	(9.46–16.93)
7–9	11.03	0.38	28.01	(23.76–32.69)	10.74	(8.00–14.24)
10–12	10.79	0.21	25.62	(23.08–28.33)	8.41	(6.88–10.25)
13–16	9.18	0.27	17.93	(14.82–21.52)	5.23	(3.58–7.56)
17 +	8.12	1.06	12.35	(5.31–25.43)	3.70	(0.70–14.36)
Marital status						
Never married	12.25	0.30	33.94	(30.12–37.98)	11.08	(8.71–13.98)
Married	9.69	0.15	21.42	(19.61–23.36)	6.90	(5.83–8.16)
Widowed	15.05	1.06	40.83	(29.67–52.99)	22.50	(13.94–34.06)
Divorced	21.86	2.40	60.47	(40.05–77.92)	39.53	(22.08–59.95)
Separated	16.70	2.32	50.00	(25.11–74.89)	25.00	(8.36–53.32)
Occupation						
High officer	8.19	1.56	17.24	(5.05–42.62)	6.90	(0.83–30.68)
Professional	8.54	0.57	16.85	(10.62–25.56)	5.06	(2.04–11.50)
Semi-professional	7.50	0.59	16.77	(10.39–25.79)	4.19	(1.48–10.64)
Office worker	10.07	0.36	23.85	(19.11–29.32)	5.65	(3.41–9.15)
Service worker	11.04	0.35	26.02	(21.88–30.64)	9.36	(6.80–12.71)
Farmer or fisherman	9.31	0.49	18.28	(13.01–25.00)	7.24	(4.08–12.40)
Licensed worker	10.33	0.40	25.52	(20.66–31.05)	7.05	(4.51–10.80)
Machine operator	12.25	0.65	33.49	(25.59–42.40)	10.23	(5.87–17.08)
Laborer	11.89	0.58	28.39	(22.22–35.47)	11.36	(7.41–16.94)
Soldier	6.98	1.26	13.04	(2.49–41.70)	4.35	(0.14–31.46)
Student	11.75	0.63	33.76	(26.15–42.29)	9.40	(5.39–15.75)
Housewife	10.09	0.29	23.23	(20.03–26.76)	8.96	(6.91–11.51)
No occupation	14.37	0.79	39.22	(31.17–47.89)	17.67	(11.97–25.23)

Table 2 (Continued)

Characteristic	CES-D score		Caseness rate (probable) ^a		Caseness rate (definite) ^b	
	Mean	S.E.	Rate	99% confidence limits	Rate	99% confidence limits
Self-assessment of his/her living standards						
Very low	15.98	1.02	41.72	(32.00–52.10)	25.77	(17.75–35.74)
Low	12.01	0.30	29.90	(26.39–33.65)	11.14	(8.86–13.90)
Average	9.94	0.16	23.46	(21.45–25.59)	7.02	(5.86–8.38)
High	8.22	0.46	17.49	(12.44–23.97)	5.61	(2.94–10.27)
Very high	14.16	3.50	27.27	(9.16–56.75)	22.73	(6.71–52.43)
Monthly income (unit: million won) ^c						
0–0.5	14.88	0.68	36.93	(30.02–44.41)	20.92	(15.43–27.67)
0.51–1.0	11.29	0.29	27.64	(24.29–31.25)	10.20	(8.06–12.82)
1.01–1.5	10.44	0.26	23.83	(20.78–27.16)	7.42	(5.67–9.64)
1.51–2.0	9.38	0.28	24.77	(21.12–28.80)	4.53	(2.98–6.81)
2.01–2.5	9.39	0.44	18.58	(13.80–24.51)	9.02	(5.75–13.80)
2.51 +	9.26	0.38	21.37	(16.93–26.58)	7.06	(4.55–10.74)
Social class						
High	8.84	0.31	19.82	(16.08–24.17)	6.46	(4.35–9.45)
Upper middle	9.77	0.23	22.30	(19.48–25.38)	5.85	(4.38–7.77)
Lower middle	10.68	0.22	24.91	(22.24–27.78)	7.86	(6.29–9.77)
Low	12.75	0.38	30.22	(26.21–34.55)	14.56	(11.63–18.08)

^aCES-D score, 16 and over^bCES-D score, 25 and over^cOne million won was equivalent to US\$ 1,218 at the time of this survey

4. Discussion

In our study, the mean CES-D score was similar to those of Japanese workers and Chinese-Americans (Ying, 1988; Iwata et al., 1989), and — except for a few studies involving Afro-Americans and Puerto Ricans (Gary, 1985; Moscicki et al., 1987; Hauenstein and Boyd, 1994) — higher than in most reports describing the situation in the US and Canada (Myers and Weissman, 1980; Barnes et al., 1988). Except for studies involving rural women (Hauenstein and Boyd, 1994) and Puerto Ricans (Moscicki et al., 1987), the prevalence rate of probable depression ($\text{CES-D} \geq 16$) was also higher among Korean adults than those in other surveys. An analysis of sample data drawn from the National Institute of Mental Health Epidemiologic Catchment Area (ECA) study indicated that 23.1% of participants met the criteria for depressive symptoms (subclinical depression) on a lifetime basis (Johnson et al., 1992). However, the

rate on the current time basis would be lower than this figure. On the basis of the finding that Korean samples scored higher on depression, certain assumptions may be made. Firstly, Korean society has experienced dramatic sociocultural changes during the last three decades, with job opportunities, family structure, and moral values all affected. The changes began with the swift westernization and industrialization of society in the late 1960s; many people left rural areas to seek new jobs in the cities, a change followed by transition from the traditional three-generation extended family structure to one that is nuclear. This has resulted in poorer emotional support from family members in times of emotional suffering (Lee, 1986). Most people have been preoccupied with getting wealthier. Competition for higher job status and living standards, and the desire not to fall behind others have caused considerable distress. In addition, many have found it difficult to adjust to a rapidly changing system of

Table 3

Logistic regression analysis of the effects of sociodemographic characteristics on definite CES-D caseness levels (≥ 25 score)

Characteristic		Coefficient	Odds ratio	P
Intercept		3.1503		
Sex	Male			
	Female	0.3825	1.4659	0.0023 ^a
Age	20–29			
	30–49	–0.1791	0.8360	0.3059
	50–59	–0.5524	0.5756	0.0295
Education	0–6	0.7171	2.0485	0.0022*
	7–12	0.5421	1.7196	0.002*
	13 +			
Marital status	Never married	0.4537	1.5741	0.0163
	Married			
	Disrupted	1.0905	2.9758	0.0001 ^a
Occupation	Employed			
	Unemployed	0.5289	1.6971	0.0133
Monthly income (million won) ^b	0–0.5	0.5211	1.6839	0.0689
	0.51–1.5	0.0348	1.0354	0.8899
	1.51–2.5	–0.3855	0.6801	0.1678
	2.51 +			

^a $P < 0.0045$ (0.05/11)^b One million won was equivalent to US\$ 1,218 at the time of this survey.

social values, from Confucian frugality to the worship of wealth. In a comparative study of affective disorders in China, Japan, and Korea, HRSD (Hamilton Depression Rating Scale) symptom scores were highest among Korean patients, and this was partly explained by the rapid social, economic and political changes taking place in Korean society (Nakane et al., 1991). Secondly, we suggest that a culturally different response style to questions relating to emotion has caused higher CES-D scores. Koreans are quite reluctant to respond positively to positive affect questions. They seem to strongly resist expressing their affect positively, and when compared to US populations, tend to choose an intermediate response to positive items (Iwata et al., 1989; Cho and Kim, 1998). This has already been pointed out with regard to studies carried out for the Japanese and the Chinese-Americans (Rankin and Galbraith, 1993; Iwata et al., 1994), and particularly because of Confucianism, Korea shares a

good deal of their common culture. Iwata et al. (1998) reported that positive affect items with positive wording cannot be used to adequately assess depressive disorders among the Japanese population, though corresponding negatively revised items are effective. In the Korean version of the CES-D used in this study, only one positive item was revised negatively. If all four had been changed, the scores in this study might have been lower. In a validating study of the Korean version of the Beck Depression Inventory, it was indicated that, because of the tendency to avoid extreme positive answers, the total score was higher than in Western countries (Hahn et al., 1986). Noh et al. (1992) reported that Korean immigrants to Canada were also more likely than whites to respond dysphorically to positive affect items. The reason for a high symptom score and high prevalence of depression among Puerto Ricans in the US is explained by the culturally determined response style of this ethnic group:

they may be more willing to report symptoms of distress (Guarnaccia et al., 1990; Cho et al., 1993). A similar explanation might apply to Koreans, who are often described as being more open, outgoing and direct in describing their emotional symptoms. It has been reported that this contributed to a depressive symptom score higher than that of Chinese and Japanese (Nakane et al., 1991).

In previous research using the CES-D scale, depression has most commonly been reported among females, younger respondents, lower socioeconomic groups, and the divorced/separated population (Comstock and Helsing, 1976; Radloff, 1977; Craig and Van Natta, 1979; Husaini et al., 1980; Madianos et al., 1988; Narrow et al., 1990), and the result of our study supported these findings. Variables related to socioeconomic status showed the highest scores among those who were divorced/separated, who had the lowest level of formal education, and who assessed their living standards as 'very low'.

As in other countries, lower rates of depressive symptoms were found among those who were highly educated and those employed in professional/managerial positions. People in these groups seem to have less difficulty in adjusting to changes in the social environment and controlling work-related distress. As in previous studies, the highest mean depression scores and highest rates of depressive symptoms occurred in the divorced/separated group, and among individuals in households with the lowest incomes; it appears that lack of emotional support by a spouse and extreme poverty lead to increased depression. The rate was higher among females, and this finding is consistent with most previous general population surveys of depression in which the CES-D scale was employed (Comstock and Helsing, 1976; Craig and Van Natta, 1978; Husaini et al., 1980; Moscicki et al., 1989). The ratio of rates of depressive symptoms between females and males in previous general population surveys employing the CES-D varies widely, from 1.25 in a Canadian urban sample (Barnes et al., 1988) to 3.24 among Cuban Americans (Moscicki et al., 1987). In this study, the female:male ratio for depressive symptoms is 1.53:1, slightly lower than the universal 2:1 ratio

for clinical depression (Johnson et al., 1992), and major depression defined by DSM-III criteria (Weissman et al., 1996), reported throughout the world. The ratio is also lower than that in a previous nationwide survey in Korea that measured the prevalence of major depression and dysthymia (Lee et al., 1990a,b; Weissman et al., 1996). In our study, the difference between the sexes was more prominent in the definite (≥ 25) than the probable depression group (≥ 16), a finding consistent with those of previous studies using the CES-D scale, in which women were more likely to fall into the severely depressed category (Hsu and Marshall, 1987; Madianos et al., 1988). An analysis of ECA data showed that the gender ratio (F/M) was 1.57:1 in a non-diagnostic depressive symptom group, and 2.47:1 in a clinical diagnostic depression (major depression and dysthymia) group (Johnson et al., 1992). Even though we used a cutoff score of 25 to capture the definite caseness of depression, the CES-D was not originally developed to generate a clinical diagnosis of depression but to detect those with depressive symptoms.

Those in their forties are in the best mood state, and the rate of depression rises again after the age of 50. The prevalence of depressive symptoms, as measured by the CES-D, for an urban Korean community sample aged 65 and over was as high as 32.5% (Cho et al., 1998). Those aged 30–49 are the most emotionally stable, and it seems they are better able to handle distress and that their jobs and marital status are the most stable. After the age of 50, many Korean employees are likely to face early retirement, and, in addition, a social security system for the elderly has not yet been established in Korea, so retirement may be directly related to financial difficulties. As found in previous studies, among several putative risk factors for depression, female gender, a low educational level (6 years and below; 12 years and below), and a disrupted marriage were independent risk factors for severe 'definite' depression ($\text{CES-D} \geq 25$) in this study. As in both Japan and Taiwan, Koreans have a burning desire to achieve a high level of education, and this, in Korea, proved to be most intimately related to higher job status and more job

Table 4

Point prevalence of depressive symptoms measured by the CES-D at cutoff scores other than 16

Author (year)	Location/sample	Cutoff score	Rates(%)		Total
			Male	Female	
Amenson and Lewinsohn (1981) ^a	Oregon, USA/community adult volunteers	18 (T1)	16.0	23.3	20.9
		18 (T2)	13.2	21.4	18.8
Murrell et al. (1983)	Kentucky, USA/urban elderly (75 yrs old and over)	20	13.7	18.2	–
Hsu and Marshall (1987)	Ontario, Canada/young physicians in training	16–20	9.7	11.6	10.4
		21–30	7.3	11.0	8.5
		31 or more	2.2	7.3	3.9
Madianos et al. (1988) ^b	Athens, Greece/community adults (age: 19–64)	16	9.2	23.3	17.6
		25	–	–	–
Garrison et al. (1991) ^c	South Carolina, USA/school adolescents (grade, 7–9)	12 for male 22 for female	–	–	–
Roberts et al. (1991)	Oregon, USA/school adolescents (grade, 9–12)	16			48.0
		21	–	–	31.4
		31			12.1
Present study (1998)	Korea/adults (age, 20–59) nationwide sample	16	23.1	27.4	25.3
		25	6.8	10.4	8.7

^aT1, Time 1 survey; T2, Time 2 survey (one year after T1).^bThe rates of those whose CES-D scores were 25 or more were not estimated.^cOnly optimal cutoff scores for male and female adolescents were presented.

security, as well as to a higher income (Nam et al., 1996). It may very well be why low educational level is a more accurate predictor of depression than is employment status or monthly income.

While the use of 25 as the caseness score in Korea was justified by previous studies (Cho and Kim, 1993), various caseness scores for the CES-D have been used or proposed as shown in Table 4. Although a cutoff point of 25 on this scale might define definite caseness of depressive symptoms, this does not specifically indicate a clinical diagnosis of depression. Thus it is difficult to compare the result of this study with that of a previous nationwide survey by Lee et al. (1990a,b; Lee et al., 1990), which used a diagnostic approach. For this, a two-stage epidemiologic survey, applying a standardized diagnostic instrument, such as the DIS is needed.

The limitation of this study lies in its cross-sectional design, which makes it difficult to determine whether a sociodemographic correlation associated with depressive symptoms is an antecedent or a consequence of depression. In order to make inferences with regard to causality, a longitudinal follow-up study is needed.

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